DATA ANALYST INTERNSHIP

Task 5: Exploratory Data Analysis (EDA)

Project: Retail Sales Data Exploration

Dataset note: This is a synthetic practice dataset created for this exercise. It contains 500 sample orders and should not be treated as real company sales records.

1. Objective and tools

The objective is to explore the dataset, check data quality, identify patterns and relationships, and communicate the findings using statistics and visualizations. The analysis uses Python, Pandas, Matplotlib and Seaborn.

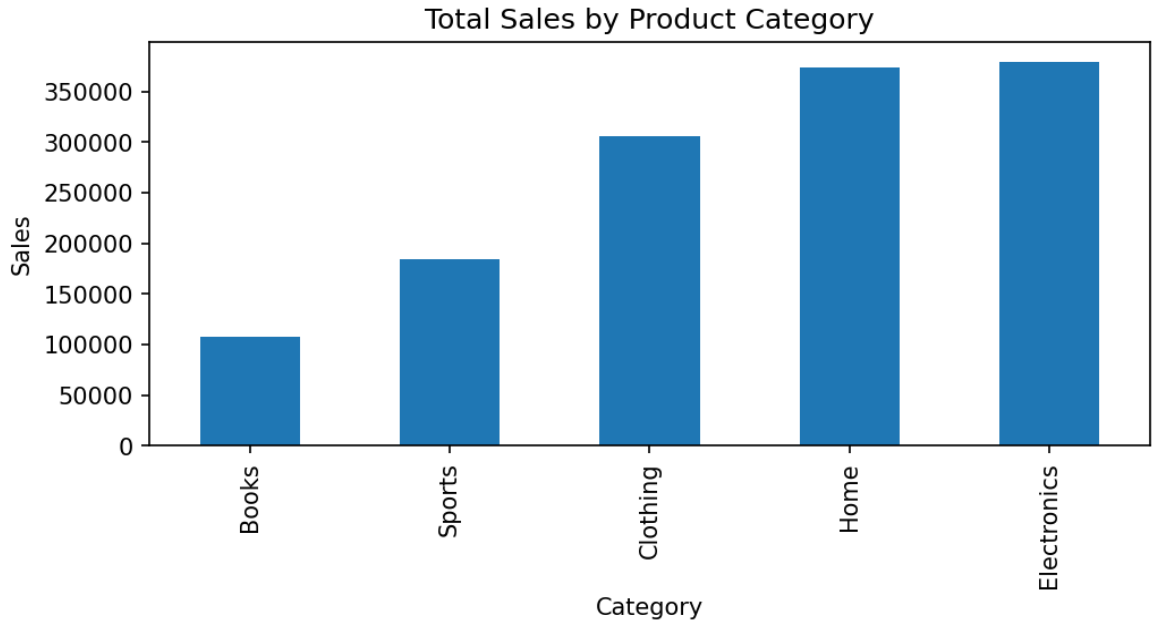
2. Dataset overview

The dataset contains 500 rows and 8 original fields plus a derived Month field during analysis. It includes order date, product category, region, units sold, unit price, discount and sales amount. Missing category and region values were filled using the mode of their respective columns.

Total sales in the sample: **1,350,886.65**. Average sales per order: **2,701.77**. The category with the highest total sales is **Electronics**, and the region with the highest total sales is **North**.

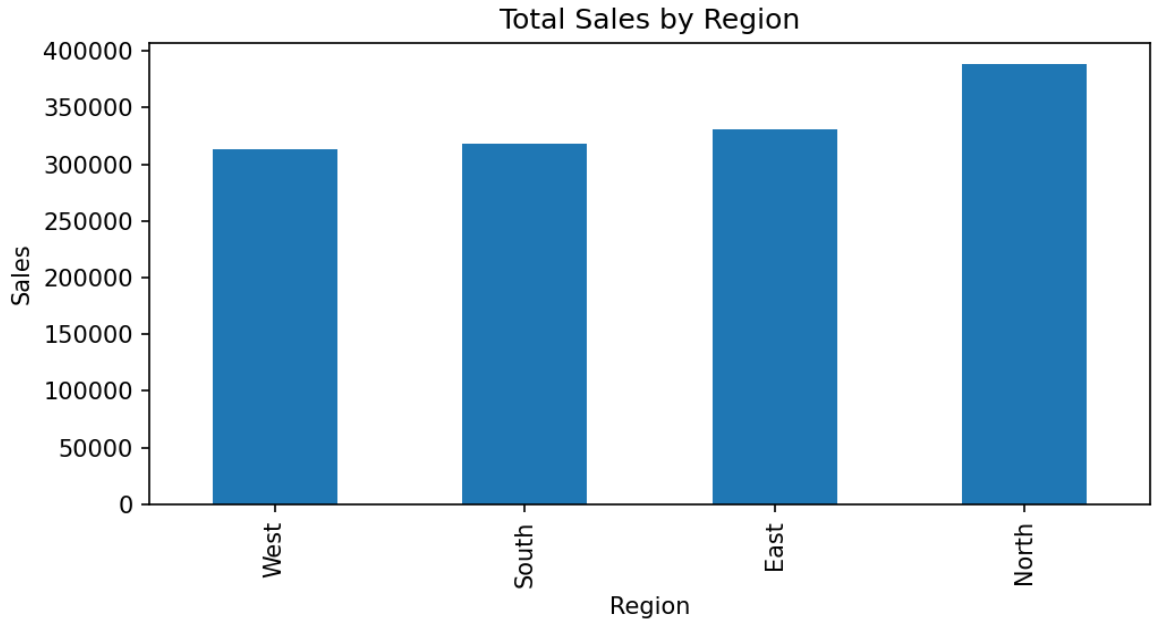
3. Visual analysis and observations

Sales by category



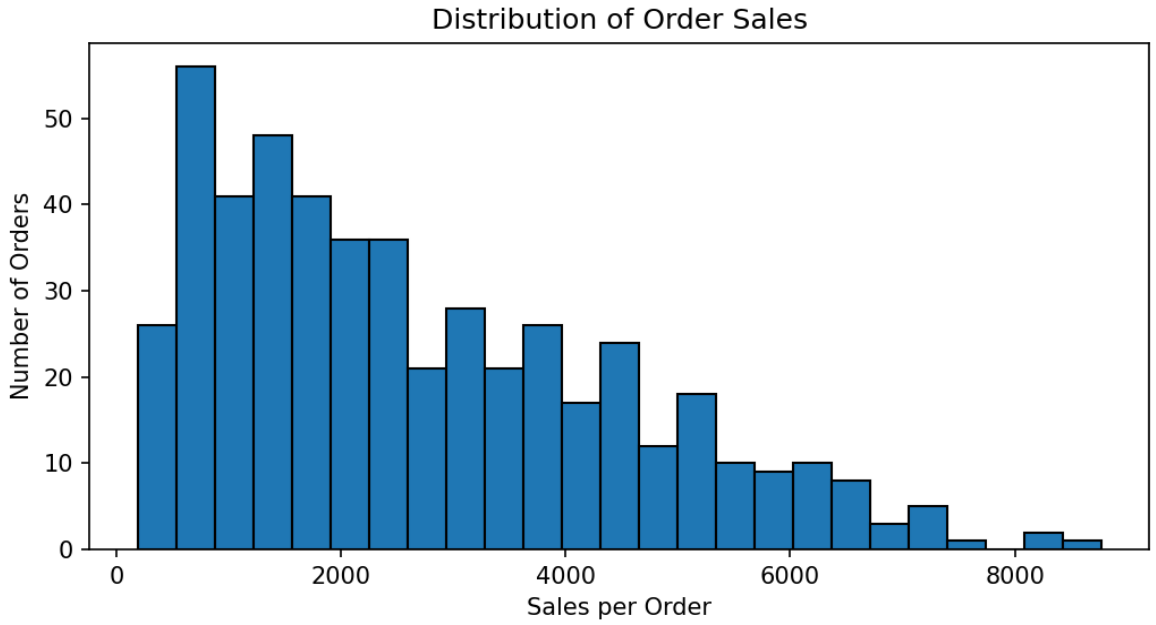
Observation: Electronics has the highest total sales among the categories in this sample.

Sales by region



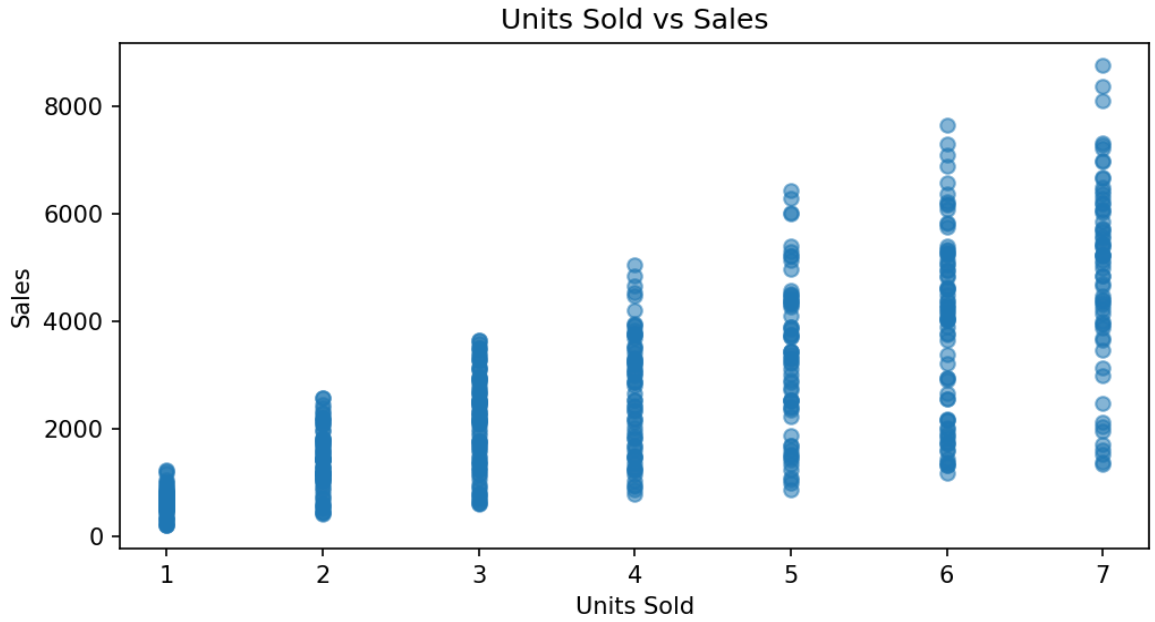
Observation: North records the highest total sales among the four regions.

Distribution of order sales



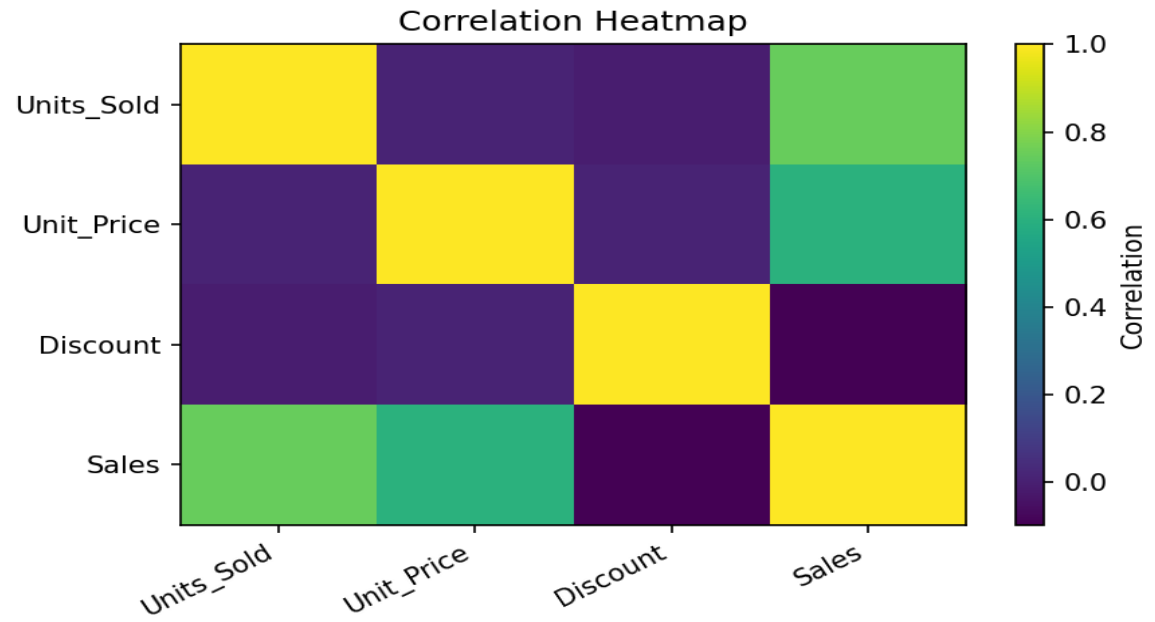
Observation: The histogram shows the spread of order values and helps identify common ranges and unusually large orders.

Units sold versus sales



Observation: The scatter plot helps inspect whether order value tends to rise as the number of units increases.

Correlation heatmap



Observation: The heatmap summarizes linear relationships among units sold, unit price, discount and sales. Correlation does not by itself prove causation.

4. Key findings

• Total sales across the 500 sample orders are 1,350,886.65.

• Average sales per order are 2,701.77.

• Electronics is the highest-selling category by total sales.

• North is the highest-selling region by total sales.

• The analysis shows how charts and summary statistics can make patterns easier to understand.

• The dataset is synthetic, so the findings are only for practice and should not be used as real business conclusions.

5. Conclusion

This EDA exercise covered data inspection, missing-value handling, descriptive statistics, category and region comparisons, a sales distribution plot, a scatter plot and correlation analysis. It provided practice in turning raw data into understandable observations. A useful next step would be to repeat the analysis with verified real sales data and investigate monthly trends in more detail.

6. Interview preparation

What is EDA?

Exploratory Data Analysis is the process of examining and visualizing data to understand its structure, patterns, relationships and unusual values.

Which plots help check correlation?

A correlation heatmap is commonly used for numerical variables. Scatter plots help inspect the relationship between two variables.

How can skewed data be handled?

First inspect the distribution. Depending on the purpose, a log transformation, robust statistics or suitable non-parametric methods may help.

How can multicollinearity be detected?

A correlation matrix can reveal highly correlated predictors. Variance Inflation Factor (VIF) is another common method.

What are univariate, bivariate and multivariate analyses?

Univariate studies one variable, bivariate studies two variables, and multivariate studies three or more variables together.

Difference between heatmap and pairplot?

A heatmap displays values using color, often for correlations. A pairplot shows pairwise plots for several numerical variables.

How should insights be summarized?

State the main pattern, support it with a statistic or chart, and mention any limitation in the data.